

Data Review — CRM Data Mart

Source: BigQuery CRM Data Mart · **Run date:** 2026-08-19 · **Scope:** full dataset, with no date filtering. Sensitive and business-specific values are excluded.

PART 1

Table-level details

Grain, row counts, columns + data types, null% and distinct values for all 7 tables.

dim_account				30 rows · 11 columns
Grain: 1 row / account_id (distinct 30 = total 30, matched)				
COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
account_id	NUMERIC	0,0%	30	Value distribution omitted
account_name	STRING	0,0%	29	Top company names omitted for privacy
account_addresses	STRING	100,0%	0	NULL: 30
account_description	STRING	0,0%	1	(empty): 30
brand_name	STRING	0,0%	1	(empty): 30
industry	STRING	0,0%	6	Categorical distribution omitted
website	STRING	0,0%	1	(empty): 30
linkedin_profile	STRING	0,0%	1	(empty): 30
company_size	STRING	0,0%	5	unknown: 14; 11-50: 10; 51-200: 4; 1-10: 1; 201-500: 1
created_at	DATETIME	0,0%	30	Date range retained
updated_at	DATETIME	0,0%	30	Date range retained

dim_contact

26.675 rows · 13 columns

Grain: 1 row / contact_key (SCD2 — distinct contact_key = distinct contact_id = 26.675, mỗi contact hiện chỉ có 1 version)

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
contact_key	STRING	0,0%	26.675	UUID, all unique
contact_id	NUMERIC	0,0%	26.675	Value distribution omitted
account_id	NUMERIC	0,0%	525	Value distribution omitted
contact_name	STRING	0,0%	21.344	Top values omitted for privacy
date_of_birth	STRING	0,0%	394	Categorical distribution omitted
facebook	STRING	10,3%	6.032	Value distribution omitted
university	STRING	0,0%	1	(empty): 26.675
location	STRING	0,0%	50	Categorical distribution omitted
job_title	STRING	0,0%	23	Categorical distribution omitted
created_at	DATETIME	0,0%	20.735	Date range retained
effective_date	DATE	0,0%	2.035	Date range retained
end_date	DATE	100,0%	0	NULL: 26.675
is_current	BOOLEAN	0,0%	1	True: 26.675

dim_pipeline

4 rows · 3 columns

Grain: 1 row / pipeline_id

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
pipeline_id	NUMERIC	0,0%	4	Value distribution omitted
pipeline_name	STRING	0,0%	4	Pipeline names omitted
pipeline_content	STRING	0,0%	2	(empty): 3; Business-specific description omitted

dim_stage

24 rows · 3 columns

Grain: 1 row / stage_id

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
stage_id	NUMERIC	0,0%	24	Value distribution omitted
pipeline_id	NUMERIC	0,0%	4	Value distribution omitted
stage_name	STRING	0,0%	18	Stage values omitted

dim_user

29 rows · 4 columns

Grain: 1 row / user_id

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
user_id	INTEGER	0,0%	29	Value distribution omitted
username	STRING	0,0%	29	all 29 values are unique
full_name	STRING	0,0%	29	all 29 values are unique
job_title	STRING	3,4%	15	Categorical distribution omitted

fact_deal

25.576 rows · 32 columns

Grain: 1 row / deal_id

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
deal_id	NUMERIC	0,0%	25.576	Value distribution omitted
contact_key	STRING	1,9%	20.727	NULL: 495; (UUID, various)
stage_id	NUMERIC	0,0%	17	Value distribution omitted
owner_user_id	NUMERIC	0,0%	17	min 2.969 / max 5.227
deal_name	STRING	0,0%	25.452	Top deal names omitted for privacy
deal_status	STRING	0,0%	3	3 distinct values
labels	STRING	99,6%	1	Business-specific values omitted

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
is_alumni	STRING	86,3%	2	Business-specific values omitted
group_registration	STRING	93,1%	1.573	Business-specific values omitted
course_selected	STRING	0,7%	43	Business-specific values omitted
learning_format	STRING	58,2%	5	NULL: 14.879; Business-specific values omitted
class_code	STRING	10,4%	1.834	NULL: 2.670; Business-specific values omitted
expectation	STRING	41,2%	14.416	Business-specific values omitted
promotion_code	STRING	85,2%	8	NULL: 21.796; Business-specific values omitted
gift	STRING	99,0%	21	NULL: 25.316; Business-specific values omitted
need_consulting	STRING	88,7%	2	Business-specific values omitted
preferred_consulting_time	STRING	61,4%	4	Business-specific values omitted
pain_point_captured	STRING	97,6%	2	Business-specific values omitted
pending_reason	STRING	98,4%	19	Business-specific values omitted
pending_reason_detail	STRING	98,8%	302	Business-specific values omitted
has_followup_plan	STRING	98,2%	2	Business-specific values omitted
next_step	STRING	99,7%	64	Business-specific values omitted
lost_reason	STRING	92,4%	17	Business-specific values omitted
lost_reason_detail	STRING	0,0%	706	Value distribution omitted
utm_source	STRING	43,3%	125	NULL: 11.074; Source values omitted
utm_medium	STRING	49,4%	143	Source values omitted
utm_content	STRING	58,5%	1.905	NULL: 14.953; Source values omitted
utm_product	STRING	92,3%	61	NULL: 23.600; Product values omitted

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
utm_person	STRING	82,9%	44	NULL: 21.215; Person identifiers omitted
created_at	DATETIME	0,0%	25.266	Date range retained
updated_at	DATETIME	0,0%	10.567	Date range retained
closed_at	DATETIME	3,4%	21.804	NULL: 857; min 2021-01-03 · max 2026-08-14

fact_deal_activity

54.404 rows · 7 columns

Grain: 1 row / activity_id

COLUMN	TYPE	NULL %	DISTINCT	DATA QUALITY NOTES
activity_id	INTEGER	0,0%	54.404	Value distribution omitted
deal_id	INTEGER	0,0%	12.674	min 1.491 / max 135.997
activity_type	STRING	0,0%	4	Activity categories omitted
activity_content	STRING	4,1%	44.271	NULL: 2.226; long automated stage descriptions
activity_owner_user_id	INTEGER	0,0%	27	min 0 / max 5.414
created_at	DATETIME	0,0%	42.081	Date range retained
updated_at	DATETIME	0,0%	42.083	Date range retained

PART 2

Foreign key relationships (FK → PK)

7 relationships based on the known ERD, each verified individually — no assumptions about additional relationships.

1. fact_deal.contact_key → dim_contact.contact_key 100,00%

FK non-null: 25.081 · Matched: 25.081 · Orphan: 0. UUID format check: 0/25.081 invalid format → no deal uses contact_id instead of contact_key.

2. fact_deal.stage_id → dim_stage.stage_id**100,00%**

FK non-null: 25.576 · Matched: 25.576 · Orphan: 0.

3. fact_deal.owner_user_id → dim_user.user_id**97,24%**

FK non-null: 25.576 · Matched: 24.869 · Orphan: 707 (2,76%), 4 distinct orphan values — users were deleted/deactivated from dim_user.

4. fact_deal_activity.deal_id → fact_deal.deal_id**100,00%**

FK non-null: 54.404 · Matched: 54.404 · Orphan: 0.

5. fact_deal_activity.activity_owner_user_id → dim_user.user_id**94,77%**

FK non-null: 54.404 · Matched: 51.556 · Orphan: 2.848 (5,23%), 9 distinct values — mostly overlaps orphan group #3, plus value 0 (suspected system sentinel).

6. dim_contact.account_id → dim_account.account_id**0,03%**

FK non-null: 26.675 · Matched: 8 · Orphan: 26.667 (99,97%), 517 distinct values. dim_account has only 30 rows — this relationship is broken and requires investigation before it can be trusted for analytics.

7. dim_stage.pipeline_id → dim_pipeline.pipeline_id**100,00%**

FK non-null: 24 · Matched: 24 · Orphan: 0.

PART 3

SCD Type 2 – dim_contact**0****Invalid current records**

contact_id values without exactly one is_current=true row

0**Contacts with multiple versions**

Out of 26,675 contact_id values

0**Overlapping version periods**

Version pairs with overlapping effective_date / end_date

Note: all three "0" values above occur because **no contact has ever generated a second version** — `end_date` is 100% NULL and `is_current` is always True. The SCD2 mechanism is correct at the schema level but has never been validated against real overlapping-version data. Re-run these two checks once the pipeline starts creating new contact versions.

PART 4

Data Quality Remediation

Actions required before the Data Mart can be treated as a trusted analytical layer.

1. Fix the broken account relationship

99.97% orphan rate : 26,667 of 26,675 non-null records in `dim_contact.account_id` do not match `dim_account.account_id`.

Example: a contact can contain an `account_id` that cannot be found in `dim_account`. A LEFT JOIN would therefore return NULL for account attributes.

Action: investigate whether the mismatch comes from different ID systems, an incomplete `dim_account`, or incorrect transformation mapping. Fix the mapping, then **re-run the FK validation** before using the relationship for analytics.

2. Standardize categorical and tracking values

Standardize values in `utm_source`, `utm_medium`, `utm_content`, `utm_product`, `utm_person`, `gift`, and `promotion_code`.

Example: Facebook, facebook, and FACEBOOK can be counted as three distinct values even though they represent the same source.

Action: apply casing and whitespace normalization such as `LOWER(TRIM(value))`, define mappings for equivalent values where necessary, and **re-run the distinct-value check** after transformation.

3. Fix and review high-null columns

High NULL rate ≠ automatically a data defect. Separate expected NULLs from fields where missing values indicate incomplete data capture.

Prioritize investigation: labels (99.6%), gift (99.0%), next_step (99.7%), pending_reason_detail (98.8%), pending_reason (98.4%), has_followup_plan (98.2%), pain_point_captured (97.6%), group_registration (93.1%), lost_reason (92.4%), need_consulting (88.7%), utm_product (92.3%), promotion_code (85.2%), and utm_person (82.9%).

Example: if pain_point_captured is expected to be completed during qualification but is 97.6% NULL, investigate the CRM/form workflow and improve upstream capture rather than filling NULLs in the Data Mart.

Do not artificially fill: end_date (100% NULL) is expected because no contact has generated a second SCD2 version.

4. Review low-value and source-system fields

Review fields that currently have **only one distinct value**, are almost entirely empty, or mainly represent raw source-system attributes.

Current candidates: account_description, brand_name, website, linkedin_profile, university, and fact_deal.labels.

Example: a column with one distinct value across the entire table provides little analytical value and may indicate an unused or poorly populated source field.

Action: decide for each candidate whether to remove it from the downstream analytical model, retain it as a documented source field, or investigate the upstream source and intended business use.

5. Add automated data-quality controls

The checks above should become **automated pipeline controls** rather than remaining a one-time manual validation.

Example: after each load, check fact_deal.owner_user_id → dim_user.user_id for orphan records and flag the load when the orphan rate exceeds a defined threshold.

Action: automate FK integrity, categorical consistency, high-null thresholds, low-value field checks, and SCD2 validation such as is_current uniqueness and overlapping effective periods.

AI-Assisted Analysis & Data Privacy

AI is used as an analytical interface, while data access is controlled at the Data Mart layer.

Privacy-by-design: Sensitive and credential-related fields are removed from the Data Mart before AI-assisted analysis. The AI can generate Python/SQL queries and execute them against the sanitized Data Mart through an **already-authenticated database connection**. The AI may access the data returned by its queries when required for analysis, but it does not receive or handle database credentials, and excluded sensitive fields are not available in the Data Mart.

Privacy workflow

CRM / Source Data → Remove sensitive & credential-related fields → Sanitized Data Mart → AI generates Python / SQL query → Authenticated database connection → Query executed → Query result returned to AI → AI analyzes and documents findings

DATA / ACCESS	AI ACCESS
Analytical data in the Data Mart	✓ Available for analysis
Query results	✓ Returned when required
Schema / metadata	✓ Available
Database credentials / secrets	✗ Never exposed
Email / phone	✗ Removed
Credential-related fields	✗ Removed
Sensitive fields excluded from the Data Mart	✗ Not available

Key principle: Privacy is enforced at the **data-access and data-model layer**, rather than relying on the AI to hide sensitive information after retrieval. The AI can work directly with the analytical data it needs, while credentials and excluded sensitive fields remain outside its access boundary.

Source: BigQuery CRM Data Mart (dim_account, dim_contact, dim_pipeline, dim_stage, dim_user, fact_deal, fact_deal_activity)

Run date: 2026-08-19 · no filtering by creation date

Privacy: sensitive, identifying, and business-specific values are excluded. Aggregate quality metrics and schema-level statistics are retained.

Public sharing: No raw customer records, credentials, or sensitive business values are included in this report.